

Hi, I'm

Juan Carlos Ruiz García

Lead AI Engineer

PhD, Computer & Telecom Engineering · AI Specialization · Cum Laude

Currently at GFT Technologies

"Turning AI into real products."

CONTACT

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- scholar.google.es/citations
- Spain · Madrid

CORE TECH STACK

GENAI & AGENTS

LangGraph · Semantic Kernel · Google ADK · Strands Agents · MCP · GraphRAG

LANGUAGES & ML

Python · Scikit-learn · Pandas · NumPy · Jupyter

CLOUD & BACKEND

Azure · GCP · FastAPI · Docker · Kubernetes · Kafka · Microservices · SaaS

AUTOMATION & DEVOPS

Terraform · Helm · Azure DevOps · n8n · GitHub Actions · GitLab CI/CD · Playwright

AI / DATA

Predictive Modeling · Feature Selection · Time Series · GenAI / LLMs · NLP · Prompt Engineering

01 PROFILE

Lead AI Engineer with a PhD in Computer & Telecom Engineering (AI specialization), bridging advanced research with production-grade AI systems. Currently architecting enterprise Agentic AI platforms (multi-agent orchestration, MCP, GraphRAG, PII guardrails). Hands-on experience in cloud architecture (Azure, GCP), process automation, and ML productization. Proven track record leading R&D initiatives funded under the prestigious NEOTEC excellence program, managing multi-tenant SaaS infrastructure, and translating business requirements into scalable technical solutions. Research background in applied ML for e-Health and HCI, plus Master's-level teaching in Big Data and Business Intelligence.

02 PROFESSIONAL EXPERIENCE

GFT Technologies — Oct 2026 – Present

Full-time · Remote · Madrid, Spain

Lead AI Engineer

- **Agentic Platform:** architecting an enterprise Agentic Platform leveraging frameworks such as LangGraph, Semantic Kernel, Google ADK and Strands Agents, alongside the Model Context Protocol (MCP).
- **Knowledge & Guardrails:** GraphRAG for knowledge retrieval and Presidio for PII guardrails.
- **Integration & Security:** event-driven messaging with Kafka / Azure Event Hubs and OAuth2-based authentication.
- **Deployment Automation:** full automation with Docker, Kubernetes (Helm), Terraform and Azure DevOps CI/CD pipelines.

JubilaME — Mar 2025 – Sep 2026

Full-time · Remote · Madrid, Spain

Lead AI Engineer — Aug 2025 – Sep 2026

- Took full ownership of the company's technological ecosystem and cloud infrastructure while continuing to drive advanced data-science initiatives.
- **Cloud Architecture:** managed and scaled GCP infrastructure for concurrent deployment of B2B platforms serving major institutional clients.
- **Backend & Microservices:** built robust microservices with FastAPI to create scalable, modular solutions across the company.
- **Business Automation:** led end-to-end automation of business processes and IT operations using n8n and AI workflows (billing, reporting).
- **IT & Access Management:** improved internal security baseline (centralized password manager + 2FA). Administered Google Workspace.
- **Technical Leadership:** audited external software providers, conducted code reviews, and translated business needs into technical execution.

OTHER

MongoDB · SQL · Java · Flask · Flet · Linux / Bash

LANGUAGES

Spanish	Native
English	Professional

EDUCATION

PhD, Computer & Telecommunication Engineering
UAM · 2021 – 2025
Specialization in AI · Cum Laude

MSc in Research & Innovation in ICT
UAM · 2019 – 2021
Award for Excellence in MSc Thesis

BSc in Computer Engineering
Universidad de Granada · 2015 – 2019

AWARDS

PhD · Cum Laude distinction
UAM · 2025

Excellence in Master's Thesis
UAM · 2021

Google Hash Code 2017
Global Rank #1437
Team "Los pollos hermanos" · Online qualification

FUNDING LED

NEOTEC 2024 — CDTI
€325,000 · 85% funding ratio
Ministry of Science, Innovation & Universities (Spain). Code SNEO-20241008.

CONFERENCE ORGANIZATION

ACM MobileHCI 2023
Athens, Greece
Local Organizer — Workshop on Advances of Mobile & Wearable Biometrics.

IEEE AVSS 2022
Madrid, Spain
Local Organizer — 18th IEEE Conference on Advanced Video & Signal-Based Surveillance. Sponsored by IEEE Computer Society PAMI TC & IEEE Signal Processing Society.

RESEARCH GROUP

BiDA Lab
Universidad Autónoma de Madrid
Biometrics & Data Pattern Analytics — a leading Spanish research group in biometrics, HCI and applied ML.

Machine Learning & Data Scientist – Marketing — Mar 2025 – Aug 2025

- **Predictive Modeling:** designed ML models for customer micro-segmentation, optimizing marketing campaigns and personalized engagement.
- **Marketing Automation:** developed data-driven workflows to maximize ROI and improve targeting efficiency.
- **AI Integration:** integrated GenAI tools and LLM prompting into marketing strategies for content generation and engagement.
- **Prototyping & Analytics:** built Python prototypes and ran A/B testing for decision-making.

UNIE Universidad — Sep 2023 – Sep 2025

Part-time · On-site · Madrid

- **Adjunct Professor — MSc Big Data & Business Intelligence (MaBDBI):** Advanced Statistics and Data Mining (regression, Bayesian learning, time series, recommender systems).
- **Adjunct Professor — BSc Computer Engineering:** user interfaces and UI prototyping.
- **Coordinator of External Internships** for MaBDBI and MSc Cybersecurity (MaCBS), Nov 2024 – Sep 2025.
- **Coordinator of the Master's Final Thesis** (MaCBS), Nov 2024 – Sep 2025.

Universidad Autónoma de Madrid — Sep 2019 – Dec 2024

Full-time · Hybrid

Research Data Scientist — AI for e-Health — Sep 2021 – Dec 2024

- Designed health interaction tools and applied ML for Healthcare.
- Developed quantitative metrics for assessing children's motor and cognitive development through tablet devices.
- Built a Fall Detection System (FDS) for elderly people using smartwatch inertial sensors.
- Authored multiple publications in IEEE Access, IEEE TETC and Pattern Recognition.

Research Assistant — Electronics & Communication Tech. — Mar 2020 – Sep 2021

- Data analysis in e-Health and e-Learning research lines.
- Joined BiDA Lab (Biometrics and Data Pattern Analytics).

Research Assistant — Computer Engineering — Sep 2019 – Mar 2020

- Data Scientist in Learning Analytics; mining of MOOC data on the edX platform.

AT A GLANCE

€325K R&D funding led (NEOTEC)

10+ peer-reviewed publications

4+ years applied ML research

2 int'l conferences organized

2 years university teaching

DOMAINS

FinTech · InsurTech · e-Health · e-Learning · HCI · Wearables · Optimization · SaaS B2B

BEYOND WORK

Football · Cycling · Padel · Hiking · Gaming · Board games · Friends

INTERNATIONAL PUBLICATIONS

1. **D. Jimenez Oviedo, R. Vera-Rodriguez, R. Tolosana, J.C. Ruiz-Garcia, J. Herreros-Rodriguez. Longitudinal Digital Phenotyping for Early Cognitive-Motor Screening.** in *Proc. IEEE Conference on Artificial Intelligence*, Granada, Spain, May 2026.
2. **J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, A. Morales, J. Fierrez, J. Ortega-Garcia and J. Herreros-Rodriguez. Longitudinal Analysis and Quantitative Assessment of Child Development through Mobile Interaction.** *IEEE Access*, 2024.
3. **J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, J. Fierrez and J. Herreros-Rodriguez. ChildCI Framework: Analysis of Motor and Cognitive Development in Children-Computer Interaction for Age Detection.** *Cognitive Systems Research*, 2024.
4. **J.C. Ruiz-Garcia, C. Hojas, R. Tolosana, R. Vera-Rodriguez, A. Morales, J. Fierrez, J. Ortega-Garcia and J. Herreros-Rodriguez. Children Age Group Detection based on Human-Computer Interaction and Time Series Analysis.** *International Journal on Document Analysis and Recognition*, 2024.
5. **J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, A. Morales, J. Fierrez, J. Ortega-Garcia and J. Herreros-Rodriguez. ChildCIdb_v2: A Longitudinal Database for Children-Computer Interaction on Mobile Devices.** in *Proc. ACM International Conference on Mobile Human-Computer Interaction Workshop, MobileHCIw*, Athens, Greece, September 2023.

Granada Dynamics — Mar 2019 – Sep 2019

Internship · On-site · Granada

Python Junior Developer

- Testing tasks and maintenance of NoSQL / MongoDB databases.
- Development of conversational agents and NLP pipelines.
- Web services and REST API development.

03 KEY PROJECTS

NEOTEC Excellence Program 2024 — Mar 2025 – Present

JubilaME × CDTI · AI-Driven Financial & Pensional Planning · €325K

Code: SNEO-20241008

Link: cdti.es/ayudas/ayudas-neotec-2024

- High-impact collaboration with CDTI under the prestigious NEOTEC program, focused on the digital transformation of the Silver Economy.
- Designed predictive models for micro-segmentation and financial behavior analysis powering a hybrid B2B / B2B2C SaaS platform.
- Deployed a scalable, multi-tenant architecture on GCP with robust encryption and GDPR-compliant identity management.
- Led technical execution for the "Bono Doctor" excellence certification, optimizing the public funding ratio to 85%.

GCP · FastAPI · Predictive Modeling · SaaS · FinTech

Gemini AI Betting Bot — Feb 2026

End-to-End Automation & Intelligent Analysis

Repo: github.com/juanka1995/apuestas-bot

- Integrated Google Gemini "Deep Research" with Python / Playwright to generate portfolios automatically.
- Orchestrated 4+ automated workflows via GitHub Actions (cron-based); containerized with Docker.
- Built a bespoke Chrome extension (Manifest V3) for automated cookie rotation and secure credential management.

Python · Playwright · GitHub Actions · Docker · LLMs

ChildCIdb — Child-Computer Interaction Database — Dec 2019 – Dec 2024

Universidad Autónoma de Madrid · BiDA Lab

Repo: github.com/BiDAlab/ChildCIdbLong

- The largest publicly available CCI dataset to date — 615+ children aged 18 months to 8 years, 2.1K+ sessions and 12.6K+ tests across 4 academic years (2019/20 – 2022/23).
- Designed acquisition protocols, mobile data collection app, and longitudinal curation pipeline.
- Foundation for 5+ peer-reviewed publications in IEEE journals.

Mobile · Longitudinal · HCI · e-Health · IEEE

6 **J.C. Ruiz-Garcia, R. Tolosana, R. Vera-Rodriguez, C. Moro. CareFall: Automatic Fall Detection through Wearable Devices and AI Methods.** in *Proc. ACM International Conference on Mobile Human-Computer Interaction Workshop, MobileHCIw*, Athens, Greece, September 2023.

7 **R. Tolosana, J.C. Ruiz-Garcia, R. Vera-Rodriguez, J. Herreros-Rodriguez, S. Romero-Tapiador, A. Morales and J. Fierrez. Child-Computer Interaction with Mobile Devices: Recent Works, New Dataset, and Age Detection.** *IEEE Transactions on Emerging Topics in Computing*, pp. 1-13, 2022.

8 **R. Tolosana, R. Vera-Rodriguez, C. Gonzalez-Garcia, J. Fierrez, A. Morales, J. Ortega-Garcia, J.C. Ruiz-Garcia, et al. SVC-onGoing: Signature Verification Competition.** *Pattern Recognition*, Vol. 127, 108609, 2022.

9 **R. Cobos, J.C. Ruiz-Garcia. Improving learner engagement in MOOCs using a learning intervention system: A research study in engineering education.** *Computer Applications in Engineering Education*. 2021; 29: 733–749.

10 **R. Tolosana, R. Vera-Rodriguez, C. Gonzalez-Garcia, J. Fierrez, S. Rengifo, A. Morales, J. Ortega-Garcia, J.C. Ruiz-Garcia et al. ICDAR 2021 Competition on On-Line Signature Verification.** in *Proc. International Conference on Document Analysis and Recognition, ICDAR*, Lausanne, Switzerland, September 2021.

CareFall — Automatic Fall Detection

May 2022 – Dec 2023

BiDA Lab × Cartronic Group · University–Enterprise collaboration

- Fall Detection System processing 3-axis accelerometer and gyroscope signals from commercial smartwatches.
- Implemented and benchmarked threshold-based (SMV, Fall Index) vs ML-based classification approaches.
- ML approach significantly outperformed traditional methods on public datasets in accuracy, sensitivity and specificity.

Wearables · Signal Processing · ML · e-Health · Smartwatch

FUTCheap — AI-Powered FIFA Squad Builder

Apr 2017 – Jan 2023

Metaheuristic optimization for Ultimate Team squads

Link: futcheap.com

- Application using metaheuristic algorithms to build optimal FIFA Ultimate Team squads under customizable constraints (price, chemistry, league, formation).
- Co-developed the foundational early phases alongside the lead developer.
- Per-position weighting, weak-foot/skills configuration, budget optimisation to maximize team quality at minimum coin cost.

AI · Optimization · Metaheuristics · Java · Python

Genetic Algorithm for Feature Selection

Jan 2021 – Apr 2022

BiDA Lab · github.com/BiDAlab/GeneticAlgorithm

- Python framework for evolutionary feature selection (tournament selection, crossover, mutation, intelligent early-stopping).
- Modular and parallelizable, compatible with any Scikit-learn estimator.
- Core component of IEEE Transactions on Emerging Topics in Computing publication.

Python · Scikit-learn · Evolutionary Algorithms · Feature Selection · IEEE